

Light beams and shafts

Light beams or sun shafts are the rays of sun light that are seen around an object such as a tree. Games such as *Crysis* and *S.T.A.L.K.E.R. Clear Sky* use this technique to great effect in their game scenes

High dynamic range lighting

Also known as HDR lighting, is a technique implemented in games, derived from photography, which allows a wider range between the lightest and darkest areas of a particular scene. In other words, scenes with excess brightness and contrast are rendered to look more realistic

Graphic Card Test

profile PCBs. For such cards, a half height metal bracket is essential, so that it can fit into HTPC cabinets. Some of these cards have scored higher in the features section.

You will also see full height PCBs in this category, ranging from the 9500GTs to even some HD4650s from ATI, which do show a range of performance results. The main reason we have kept a wide first category is because we are targeting those users who want HTPCs and at the same time need a graphics card on a budget.

The chipsets that we tested in this category include NVIDIA's G98 (8400GS), G96 (9400GT and

9500GT), G84 (8600GT) and G92 (9600GSO), while AMD's contribution in this category includes the RV710 (HD 4350) and RV730 (HD 4650).

From the table, you will see a big performance difference between the lower-end 8400GS, 4350 and the 9600GT and HD4670, which clearly shows what you will get by increasing your budget by Rs. 1,000 or Rs. 1,500.

Features

The factors that are important for graphic cards for HTPC are size, connectivity, noise and temperature. While noise and temperature is something that is important for both categories, it is more important here for the simple reason that these cards will have to survive the gruelling temperature levels within a smaller HTPC cabinet. You also don't want the machine playing your videos to drown out the soundtrack.

DVI, D-Sub and S-Video

are the standard connects located on most graphic cards, but important in this category is the inclusion of an HDMI port for connectivity to large screen displays that don't come with DVI ports for digital connectivity. The cards in this category that did come with an integrated HDMI port are the Palit GeForce 9400GT 1GB, the ASUS EN9400GT, the MSI GeForce 9500GT 512MB, ASUS EAH4550 and the ASUS EAH4650.

The ZOTAC 9400GT Zone and the 9500GT Zone were almost identical in terms of



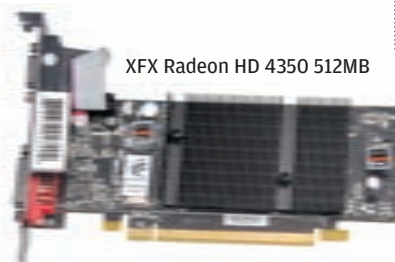
MSI NX8600GT 1GB

design and were both passively cooled with the heatsink stretching over the height of the PCB. The ASUS EN8400GS is also passively cooled, but unlike it's 8400GS low profile competitors, this one was full height. Not only that, but the heatsink extended lengthwise more than the PCB. The XFX Radeon HD 4350 is a low profile card and is passively cooled with slim heatsinks on both sides of the PCB.

We received two 8600GTs from MSI, one of which wasn't working. The one that did work, the NX8600GT, is also passively cooled with a very interesting design. The



MSI N9400GT 512MB



XFX Radeon HD 4350 512MB

HOW WE TESTED

The test rig used was:

Motherboard: ASUS P5Q3 Deluxe; based on Intel's P45 chipset this is one of the best performing P45 motherboards available.

CPU: Intel Core 2 Quad QX9650 (3.0 GHz)

RAM: 2 x 2 GB DDR3 Corsair XMS3 (1333 MHz @ 8-8-8-20)

Hard drive: 500 GB Seagate Barracuda 7200.11

Operating System: Windows Vista SP1 32-bit

Benchmarks

3D Mark 06: A theoretical test for DX9 performance.

3D Mark Vantage: FutureMark's DX 10 benchmark

Unreal Tournament III: Unreal Tournament III has been built on the Third Generation Unreal Engine and has been designed for DX9 and DX10 based systems. This game supports HDR (High Dynamic Range), dynamic shadows and per-pixel lighting which makes it a good test for graphic cards. We tested this game at 1280 x 1024 and 1600 x 1200 with

Hardware PhysX and Motion Blur enabled.

Prey: Prey is one of the few OpenGL based games based on the Doom 3 engine still doing rounds. This engine scales very well and makes a good benchmark particularly for lower-end cards. The resolutions we tested at were: 1024 x 768 and 1280 x 1024 with high settings.

Tom Clancy's H.A.W.X: This game is a great air-combat, dog fight based game of which the game engine has been developed in house by Ubisoft. We tested H.A.W.X in DX10 mode and this is one of those games that are great for multi-core CPUs. H.A.W.X. was tested with 2x AA and High settings at 1024 x 768 and 1280 x 768

Company of Heroes: Tales of Valor: The latest in the Company of Heroes series and is based on Relic's Essence Engine which features HDR, dynamic shadows and advanced shader effects, and makes for a very good test of graphic cards through all categories. We ran the game at

1024 x 768, with High setting, but shader quality was set at Medium. At 1280 x 1024, the high settings were maintained, but we increased the shader quality to Ultra.

Crysis: Crysis is a famous First Person Shooter developed on the CryEngine 2 by Crytek and stresses the GPU to the max. Lower end cards had a real rough time through this benchmark. The CryEngine2 supports Real Time Lighting and Dynamic Soft Shadows, Volumetric, Layer and View Distance Fogging, Subsurface scattering, HDR Lighting, Light shafts, Depth of Field and a whole lot more. Crysis was tested at 1024 x 768 with low settings and no AA and also at 1280 x 1024 with the same settings.

Crysis Warhead: The second installment in the Crysis series has also been developed by the same CryEngine 2, but as the FPS results show, do stress the graphic cards even further. Crysis Warhead was also benchmarked with the same resolution and settings

as Crysis.

S.T.A.L.K.E.R. Clear Sky:

The prequel to S.T.A.L.K.E.R. Shadow of Chernobyl and uses the X-Ray 1.5 game engine with features such as volumetric lighting, smoke and fire, soft water, dynamic wet surfaces, depth of field, blur and enhanced weather effects. The resolutions selected for this game were 1024 x 768 with Enhanced Full Dynamic Lighting, DX10 with High settings. We also tested this game at 1280 x 1024 resolution with the same lighting, but at low settings.

Temperature: We measured the temperature of each graphic card under idle and 100 per cent load by using Speedfan and Riva Tuner. We had to use two different programs to record the temperature because Speedfan could not record the AMD GPU temperatures while Riva Tuner couldn't record NVIDIA GPUs. To get the cards to their maximum load, we ran the S.T.A.L.K.E.R. Clear Sky benchmark in a five time loop and recorded the maximum temperature.